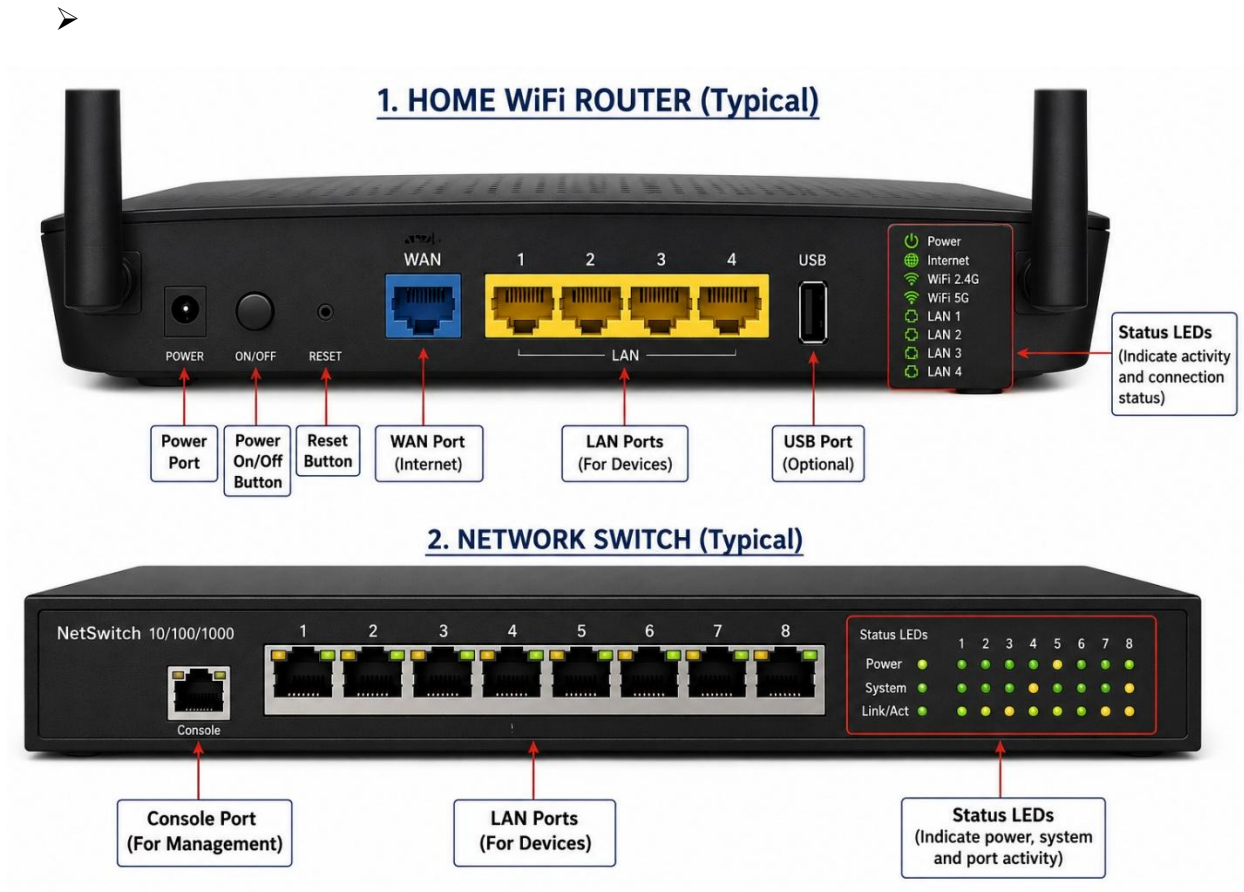


Assignment Day – 2

1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.



Feature	Router	Switch
1. Main Function	Connects different networks and directs data between them.	Connects devices within the same network and forwards data between them.
2. Address Used	Mainly uses IP addresses to decide where packets should go.	Mainly uses MAC addresses to forward Ethernet frames to the correct device.
3. Network Connection	Connects different networks, such as a home LAN to the Internet (WAN).	Connects multiple devices such as PCs, printers, and servers within a LAN.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Router

Component	Position	Purpose
Power Port	Back side, left side	Connects the router to the power adapter.
WAN Ethernet Port	Back side, near the center	Connects the router to the modem/Internet connection.
LAN Ethernet Ports	Back side, next to the WAN port	Connect computers, printers, or other wired devices.
LED Indicators	Front/top panel	Show power, Internet, Wi-Fi, and LAN connection/activity status.

Network Switch

Component	Position	Purpose
Power Port	Usually on the back or side	Supplies electrical power to the switch.
Ethernet Port 1	Front panel	Connects a computer or other network device.
Ethernet Port 2	Front panel	Connects another computer or network device.
LED Indicators	Front panel, usually above/beside Ethernet ports	Show power, link, and data/activity status.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.



Small Gaming Cafe Network

I would connect the router first to the ISP line because the router receives the Internet connection and manages communication between the café's local network and the Internet.

The setup would be:

ISP Line → Router → Network Switch → 8 Gaming PCs

The router's WAN port would connect to the ISP/modem. Then, one LAN port of the router would connect to the network switch. The switch would then connect to all 8 gaming PCs using Ethernet cables.

Justification:

- Router: Provides Internet access, assigns IP addresses (DHCP), and can provide basic firewall/NAT protection.
- Switch: Connects all 8 gaming PCs within the same LAN and efficiently forwards network traffic using MAC addresses.
- Ethernet connections: Wired connections are preferred for gaming because they generally provide more stable and lower-latency connectivity than Wi-Fi.

This setup allows the PCs to stream IPL matches and play multiplayer games while sharing the same Internet connection.